MohannadHendi at AraGenre 2026: Contrastive Cross-Lingual Definition Retrieval for Hierarchical Arabic Genre Classification

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Abstract

We describe the MohannadHendi submission to the AraGenre 2026 shared task. The system casts hierarchical Arabic genre classification as cross-lingual retrieval between Arabic texts and English genre definitions, allowing it to consider labels unseen during training. We fine-tune multilingual E5-large on 140 texts paired with five deterministic views of their genre definitions. At inference, the nearest normalized definition embedding supplies both the broad and specific labels. The best local configuration achieved 0.9020 specific Macro F1. The submitted system obtained 0.4560 Hierarchical Macro F1 and ranked eighth among 18 final leaderboard entries. Its broad Macro F1 (0.6125) exceeded its specific Macro F1 (0.2995), showing that fine-grained genre discrimination remains the main difficulty.

1 Introduction

Genre classification identifies a text's communicative purpose and organization rather than only its topic. Two texts may discuss the same event while functioning as a news report, an opinion, or a piece of advice. AraGenre 2026 frames this distinction hierarchically: every Arabic text receives a broad communicative category and a more specific genre subtype (El-Haj et al., 2026). The task combines a small labelled training set, Arabic inputs, English category definitions, and genres that may be unseen during training. The hidden evaluation data further include naturally occurring Modern Standard Arabic, Classical Arabic, and dialectal texts, creating a substantial shift from the controlled examples released for development.

A conventional classifier learns a fixed output layer whose classes are determined during training. Such a model cannot naturally score a new category from its description alone. We therefore formulate classification as cross-lingual semantic retrieval. Multilingual E5-large (Wang et al., 2024) places Arabic texts and English definitions in a shared embedding space. We construct five textual views from each broad and specific genre record and use them as positive passages during contrastive fine-tuning. At inference, the model compares a text with all candidate definitions and returns the two labels associated with the closest record.

The official development submission scored 0.8938 Hierarchical Macro F1. On final evaluation, the system scored 0.4560 and ranked eighth among 18 entries. Our contribution is a simple definition-driven system that (i) supports unseen categories without changing an output layer, (ii) constructs five deterministic training views from the genre fields supplied by the task organizers, and (iii) requires no external training examples or generative model. Code is available online.[1]

2 Background

2.1 Task Setting

Each task instance contains an identifier and an Arabic text segment. A valid prediction must use the exact broad and specific labels supplied with the candidate definitions. The training split contains 140 controlled texts spanning seven specific and five broad genres, with 20 examples for every specific genre. The development split contains 110 texts spanning six specific and six broad genres. None of its specific genres occurs in training, so development evaluation directly tests whether the model can use an unseen label description.

Final evaluation uses 27,972 naturally occurring texts and 74 candidate definition records. Compared with the released splits, it introduces wider variation in length, dialect, formality, orthography, and topic. The official Hierarchical Macro F1 averages broad and specific Macro F1 (El-Haj et al., 2026). This metric rewards systems that preserve

[1] Code repository.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 266–270
October 28-29, 2026 ©2026 Association for Computational Linguistics

the hierarchy while still distinguishing fine-grained subtypes.

2.2 Related Work

Zero-shot text classification aims to assign labels that lack task-specific training examples. Natural-language label descriptions provide a practical alternative to treating labels as arbitrary identifiers. Yin et al. (2019) formulate zero-shot classification as textual entailment, while Gao et al. (2023) show that explicit label-description training can improve robustness across domains. In hierarchical classification, HierPrompt enriches label prototypes with generated examples and incorporates the taxonomy into similarity scoring (Zhang et al., 2025). Our setting differs in that the definitions are supplied by the task and we do not use an additional language model to expand them.

The Arabic–English mismatch makes cross-lingual representation learning central to the task. Large multilingual encoders such as XLM-R demonstrate that shared pre-training can support transfer across many languages (Conneau et al., 2020). LaBSE further shows that multilingual sentences can be aligned for semantic retrieval using a dual-encoder objective (Feng et al., 2022). Multilingual E5 extends this direction with retrieval-oriented training and explicit query and passage prefixes (Wang et al., 2024). These properties motivate our use of one encoder for both Arabic inputs and English definitions.

Our prediction rule is also related to dense retrieval, where independently encoded queries and passages are compared in a common vector space. Dense Passage Retrieval showed that a dual encoder can learn effective semantic matching from paired examples (Karpukhin et al., 2020), and Sentence-BERT provides an efficient framework for similarity search over sentence embeddings (Reimers and Gurevych, 2019). We adapt this idea from retrieving evidence passages to retrieving a genre-definition record.

Distribution shift remains important in small shared-task datasets. In an Arabic-script task on AI-generated Urdu detection, Hendi (2026) found that dataset characteristics and generator artifacts affected generalization. Outside NLP, ResAF-Net adapts pretrained visual representations using feature fusion and multi-head attention for satellite-based tree detection in a resource-constrained Palestinian setting (Al-Qasem, 2026). Despite the different modality, it shares our broader strategy of adapting pretrained representations to a specialized low-resource task.

3 System Overview

The system uses `intfloat/multilingual-e5-large` as a shared-weight bi-encoder for Arabic texts and English descriptions. Both sides are processed by the same parameters rather than by language-specific encoders. This reduces the number of trainable components and preserves the multilingual alignment already learned by E5. Figure 1 summarizes the core contrastive training pipeline.

3.1 Definition Views and Training

Each genre record contains a broad name, broad definition, specific name, and specific definition. From these fields we create five views. The first joins all fields in broad-to-specific order and exposes the full hierarchy. The second keeps only the specific name and definition, encouraging attention to fine-grained cues. The third keeps the broad fields and emphasizes the general communicative function. The fourth reverses the full description to reduce dependence on a fixed field order. The final view contains only the two genre names, retaining a short lexical anchor when definitions are verbose or partially overlapping.

The views are deterministic and introduce no external facts. Pairing every one of the 140 Arabic texts with all five views of its correct genre produces 700 positive pairs. Following the E5 format, Arabic inputs use the `query:` prefix and English views use `passage:`. The prefix informs the encoder which representation plays the retrieval-query role and which acts as a candidate passage.

We fine-tune the encoder with Multiple Negatives Ranking Loss in SentenceTransformers (Reimers and Gurevych, 2019). Within a mini-batch, the genre view paired with a text is its positive passage, while passages paired with other texts act as negatives. The objective rewards a larger similarity for the matched pair relative to the remaining passages. This avoids training a fixed classifier and directly optimizes the similarity function later used for prediction.

These views are deterministic reformulations of the supplied genre fields rather than independently authored definitions. We did not compare this construction against training with only the canonical

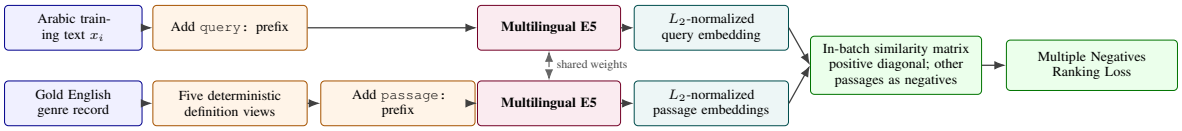


Figure 1: Contrastive training architecture. Each Arabic text is paired with five deterministic views of its gold English genre record. A shared multilingual E5 encoder produces normalized query and passage embeddings, which are optimized using Multiple Negatives Ranking Loss.

complete definition. Therefore, we treat the five views as a component of the training design and do not attribute a separate performance improvement to them.

3.2 Retrieval

At inference, we construct the complete broad-to-specific view for every candidate record. Candidate embeddings are computed once and normalized, after which each Arabic text is encoded independently. Let x denote an input text and d_j a candidate definition. We define $q(x) = \texttt{query: } x$ and $p(d_j) = \texttt{passage: } d_j$. Using the L_2-normalized encoder $\hat{f}_\theta$, the selected candidate is

$$j^* = \arg\max_j \; \hat{f}_\theta(q(x))^\top \, \hat{f}_\theta(p(d_j)) \,. \qquad (1)$$

The dot product is therefore equivalent to cosine similarity. We select the highest-scoring candidate and copy both labels from the same record. This guarantees a valid broad–specific combination even though the model has no separate classification head or hierarchy-aware loss. The candidate inventory can also change between splits without retraining an output layer.

4 Experimental Setup

We used the released training split for fine-tuning and the development split for checkpoint comparison and model selection. The hidden test split was used only by the official evaluation server. No external examples, machine translation, stemming, dialect normalization, or manual removal of diacritics was applied. Apart from adding E5 prefixes, texts were retained in their released form and truncated to a maximum of 512 tokens.

Local selection used specific Macro F1 because all development specific genres were unseen and this metric exposes performance on the more difficult level. Accuracy was recorded as a complementary instance-level measure. For official reporting, we use Hierarchical Macro F1 together with broad and specific results. Local search scores and official submission scores are kept separate because they originate from different evaluation records.

An exhaustive $3 \times 2 \times 2 \times 2$ search evaluated 24 combinations: loss scale $\{15, 20, 25\}$, epochs $\{8, 12\}$, learning rate $\{5 \times 10^{-6}, 10^{-5}\}$, and batch size $\{16, 24\}$. Fixed settings were a 0.15 warmup ratio, seed 42, mixed precision, and inference batch size 32. Experiments used `sentence-transformers` 2.7.0 in a Kaggle environment with one NVIDIA Tesla T4 GPU. Each configuration was run once, so the search measures configuration sensitivity rather than variance across random seeds.

5 Results and Analysis

Run 3 performed best with scale 15, eight epochs, learning rate 10^{-5}, and batch size 16. Across the grid, specific Macro F1 ranged from 0.8164 to 0.9020. Averaged over the remaining settings, eight epochs scored 0.8616 compared with 0.8522 for twelve epochs, suggesting that longer fine-tuning was not consistently useful on the small dataset. The two learning rates were close on average (0.8550 and 0.8587), as were batches 16 and 24 (0.8545 and 0.8592). Loss-scale averages also differed by less than 0.006. The best result therefore reflects an interaction among settings rather than a single dominant hyperparameter. Because the development set has 110 texts and every configuration was run once, these comparisons should not be interpreted as statistically stable effects.

The official development entry achieved 0.8938 Hierarchical Macro F1 and ranked 19th among 25 leaderboard entries. This value is slightly below the best local specific score of 0.9020; we report the two separately rather than treating the local run as an official result. Final evaluation produced 0.4560 Hierarchical Macro F1 and ranked the system eighth among 18 entries. Broad weighted F1 and accuracy were 0.7025 and 0.7060, whereas specific weighted F1 and accuracy were 0.3161 and 0.3196.

Runs 1–12							Runs 13–24						
Run	Scale	Ep.	LR	Batch	F1	Acc.	Run	Scale	Ep.	LR	Batch	F1	Acc.
1	15	8	5e–6	16	.8433	.8364	13	20	12	5e–6	16	.8334	.8364
2	15	8	5e–6	24	.8676	.8727	14	20	12	5e–6	24	.8838	.8909
3	**15**	**8**	**1e–5**	**16**	**.9020**	**.9091**	15	20	12	1e–5	16	.8612	.8545
4	15	8	1e–5	24	.8571	.8545	16	20	12	1e–5	24	.8164	.8273
5	15	12	5e–6	16	.8334	.8364	17	25	8	5e–6	16	.8427	.8455
6	15	12	5e–6	24	.8927	.9000	18	25	8	5e–6	24	.8523	.8545
7	15	12	1e–5	16	.8479	.8455	19	25	8	1e–5	16	.8767	.8818
8	15	12	1e–5	24	.8182	.8273	20	25	8	1e–5	24	.8623	.8545
9	20	8	5e–6	16	.8238	.8273	21	25	12	5e–6	16	.8349	.8364
10	20	8	5e–6	24	.8599	.8636	22	25	12	5e–6	24	.8927	.9000
11	20	8	1e–5	16	.8938	.9000	23	25	12	1e–5	16	.8612	.8545
12	20	8	1e–5	24	.8571	.8545	24	25	12	1e–5	24	.8501	.8545

Table 1: All local development runs. F1 is specific Macro F1; the best run is bold.

Evaluation	B-MF1	S-MF1	H-MF1	S-Acc.
Official dev.	.8938	.8938	.8938	.9000
Final	.6125	.2995	.4560	.3196

Table 2: Official broad (B), specific (S), and hierarchical (H) results.

The broad Macro F1 of 0.6125 substantially exceeded the specific Macro F1 of 0.2995. The model therefore retained useful information about general communicative purpose even when it chose the wrong subtype. The decline from development to final evaluation is consistent with the shift from short controlled examples to natural texts with more varied language and genre boundaries. It also suggests that similarity to one full definition is often insufficient for resolving closely related subtypes.

The best local model made ten development errors. Four Motivational Writing texts were classified as Advice Columns. For instance, one text encourages the reader to divide a large problem into smaller, manageable steps. Although its gold label is Motivational Writing, the model predicted Advice Columns, probably because of its direct second-person recommendation. Three Educational Explanations were classified as Analytical Reports. One such text explains gross domestic product using the equation $\mathrm{GDP} = C + I + G + (X - M)$. Its formal economic terminology and quantitative notation may have caused the model to favor Analytical Reports over the gold label, Educational Explanations. Diplomatic Communication was also mapped to Analytical Reports twice, while one Religious Commentary text was mapped to Advice Columns. Overall, these errors suggest that surface wording can sometimes outweigh the intended communicative purpose.

5.1 Discussion

Definition retrieval provides two practical advantages in this task. First, the prediction inventory is determined by the definitions available at inference rather than by a fixed training label set. Second, the selected record offers a simple semantic trace of the decision because its text can be inspected alongside the query. However, retrieving one complete record also couples the broad and specific decisions. This preserves hierarchical validity, but a strong match to the broad description can outweigh weaker evidence for the specific subtype. The training views expose broad and specific information in separate formulations, while inference compares each text with one complete representation per candidate. Since we did not conduct a view-level ablation, the effect of these alternative formulations cannot be isolated from the rest of the system. A hierarchy-aware extension could score broad definitions first and use their scores as priors when ranking compatible specific definitions. Such a design would retain support for unseen labels while allowing the two levels to contribute separately.

6 Conclusion

We presented a cross-lingual retrieval system for definition-guided Arabic genre classification. We constructed five deterministic definition views for contrastive fine-tuning on 140 texts, while nearest-definition retrieval allowed candidate labels to be scored without a fixed classification layer. The system ranked eighth on final evaluation, but the large broad–specific gap shows that future work should model the hierarchy explicitly and improve discrimination among related subtypes.

Limitations

This study uses one small development set and one seed, while reusing development data for selection and analysis. We report no view-level ablation, and related descriptions may be imperfect negatives. Final labels were unavailable for error analysis. Future work should address these constraints with multiple seeds, genre-aware negatives, and hierarchical reranking.

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